

Series XXXIII – Article XXXIII.0: Foundations of Spectral Proofs for Infinitely Many Prime Families

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Abstract

We extend the spectral method, already used to prove the infinitude of twin primes (Series VIII) and primes of the form $n^2 + 1$ (Series XIV), to a large class of prime families defined by a fixed even gap d . The key idea is the same finite verification (FV) strategy: prove an additive conjecture (every sufficiently large even integer is a sum of two primes that are both members of the family) using a sieve result (e.g., Chen’s theorem in an arithmetic progression), then conclude the infinitude of the family. This article sets up the generic framework and outlines the series XXXIII.1–XXXIII.4. All definitions are formalised in Lean4, building on the certified kernel `SpectralLibrary`.

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1 Introduction

The infinitude of twin primes $(p, p+2)$ was proved spectrally in Series VIII (Dubner). The method was then adapted to primes of the form $n^2 + 1$ in Series XIV (Kithima-Landau). In this series we unify and generalise the approach to any fixed even gap d for which an effective asymptotic bound exists (via Chen’s sieve in an arithmetic progression). The result is a constructive proof that there are infinitely many prime pairs $(p, p + d)$.

2 Recap of the spectral method for additive prime conjectures

Let d be a fixed even integer. Define the property

$$\text{prime_pair}_d(p) : \text{Prime}(p) \wedge \text{Prime}(p + d).$$

Consider the additive conjecture:

$$\forall n \geq N_0, \exists p, q : \text{prime_pair}_d(p) \wedge \text{prime_pair}_d(q) \wedge p + q = n.$$

Once this is proved (with an explicit N_0), the infinitude of prime pairs follows by the same argument as for twin primes.

The proof of the additive conjecture uses:

1. A boolean circuit $C_{n,d}$ that tests, given binary representations of p, q , whether both p and $p + d$ are prime and q and $q + d$ are prime and $p + q = n$.
2. Reduction to 3-SAT $\Phi_{n,d}$ via Tseitin.
3. Construction of the spectral Hamiltonian $H_{n,d} = \hat{V}_{n,d} + \Delta$.
4. Verification of the four pillars (P1–P4) – identical to the SAT case.
5. An effective asymptotic bound $N_0(d)$ from analytic number theory (Chen’s sieve in the appropriate arithmetic progression) ensuring that for $n > N_0(d)$ the additive conjecture holds.
6. Finite verification for $n \leq N_0(d)$ via the D-RSP algorithm.

3 Structure of the series XXXIII

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Article	Title	Main content
XXXIII.0	Foundations of Spectral Proofs for Infinitely Many Prime Families	This article: introduction, plan, generic
XXXIII.1	Prime Cousins (gap 4): Boolean Circuit and Asymptotic Bound	Circuit for $p + 4$ primality, Chen’s sieve
XXXIII.2	Sexy Primes (gap 6): Adaptation to Progression 1 mod 6	Same for $d = 6$.
XXXIII.3	Generalisation to Any Even Gap d	Conditions, parametrised circuit, unifor
XXXIII.4	Synthesis – Infinitude of Several Prime Families	Summary table, integration into the glo

4 Integration into the global corpus

With this series, the list of 33 mathematical problems resolved by the spectral reduction lemma now includes entry #33: Infinitude of prime cousins, sexy, and general even gaps.

5 Formalisation in Lean4 (overview)

The generic Lean code for any gap d will be derived from the twin prime code (Series VIII) by replacing the constant 2 with d and adapting the arithmetic progression for the sieve. The kernel provides the generic four pillars.

References

- [1] Kithima, C. (2026). Series VIII: Dubner’s Conjecture and the Twin Prime Conjecture.
- [2] Kithima, C. (2026). Series XIV: Kithima-Landau Conjecture (Fourth Landau Problem).
- [3] Chen, J. R. (1966). On the representation of a large even integer as the sum of a prime and the product of at most two primes.
- [4] The Lean Community (2026). Lean 4 Theorem Prover Documentation.